

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT		Revision No. 02 Page 1 of 10

ISSUE HISTORY			
Issue	Description of Change	Originator	Effective Date
2	Initial Release Revised as per 2015 standard on March 16, 2017 May 05, 2026	Mulugeta Tadesse DCIC DCIC	06/05/2016

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT TITLE
ISO 14001:2015 clause 6.1.2	Environmental aspect
Environmental Impact Assessment Proclamation No. 299/2002	Impact assessment

CONTENTS	PAGE
A ISSUE HISTORY	1
B REFERENCE DOCUMENT	1
C CONTENT	1
1. INTRODUCTION	2
2. PURPOSE & SCOPE	2
3. DEFINITIONS	2
4. CRITERION	3-9
4.1 MANAGEMENT RESPONSIBILITIES 4.2 EMPLOYEE RESPONSIBILITIES 4.3 KEY OPERATIONAL ENVIRONMENTAL CONTROLS 4.4 POLLUTION CONTROL 4.5 SITE WASTE MANAGEMENT PLANS 4.6 ENERGY AND RESOURCE MANAGEMENT 4.7 MATERIALS STORAGE 4.8 SUSTAINABLE SOLUTIONS 4.9 ENVIRONMENTAL INCIDENT AND EMERGENCY CONTROL 4.10 TRAINING 4.11 MONITORING 4.12 PROCUREMENT	
5. RELATED DOCUMENTS Summary of Controls to Contain Spillages to Land And Water	10
6. DISTRIBUTION	10

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT	Revision No. 02	Page 2 of 10

1. INTRODUCTION

RAMA is deeply committed to the protection, preservation, and enhancement of the environment. Our Health, Safety, and Environment (HSE) Policy Statement binds us to a core operational philosophy: *“Work collaboratively with our clients, contractors, and suppliers to improve the efficiency of our operations, conserve natural resources, and aggressively reduce waste and emissions.”*

As modern regulatory frameworks and global expectations shift, governments increasingly demand that industries actively champion **Sustainable Development**. RAMA embraces this challenge, ensuring that our economic and construction developments operate hand-in-hand with robust environmental protections to safeguard the quality of life for future generations.

2. PURPOSE AND SCOPE

This guidance document establishes the core environmental principles and operational procedures governing RAMA’s activities. It acts as the backbone of the **RAMA Environmental Management System (EMS)**.

Standard Compliance: The RAMA EMS is fully structured, maintained, and registered under the international standard **ISO 14001:2015**. This framework applies unconditionally to all corporate facilities, logistics hubs, and active construction project sites.

3. DEFINITIONS

To ensure operational clarity across all levels of management and field staff, the following ISO-aligned definitions apply:

- **Environment:** The surroundings in which RAMA operates, including air, water, land, natural resources, flora, fauna, humans, and their complex interrelations.
- **Environmental Aspect:** An element or byproduct of RAMA’s construction activities, products, or services that interacts or can interact with the environment. (*e.g., fuel storage, dust generation*).
- **Environmental Impact:** Any change to the environment, whether adverse or beneficial, resulting wholly or partially from RAMA’s environmental aspects. (*e.g., soil contamination, air quality reduction*).
- **Environmental Objective:** An overall environmental goal established by the company, perfectly consistent with our overarching HSE Policy.
- **Environmental Target:** A detailed, measurable performance requirement derived directly from our Environmental Objectives, which must be systematically met across specific projects or timelines.

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT	Revision No. 02	Page 3 of 10

4. CRITERION

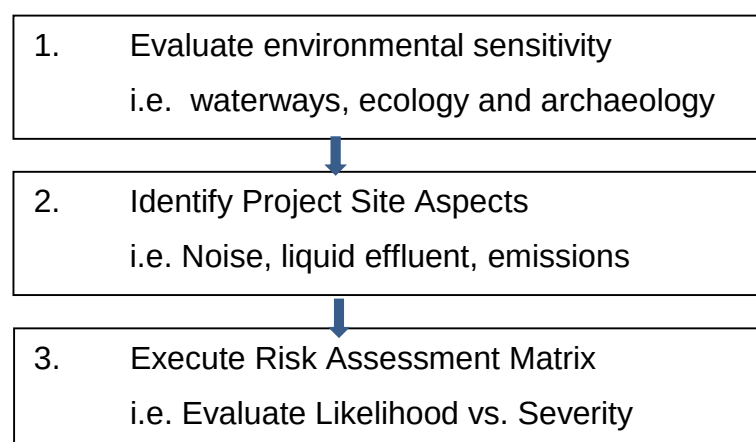
4.1 MANAGEMENT RESPONSIBILITIES

Managing Director (MD): Holds ultimate corporate accountability for RAMA's overall environmental performance. The MD champions sustainable development principles, represents the company at key stakeholder forums, and ensures the necessary resource allocation to uphold our environmental integrity.

HSE Manager: Serves as the primary technical advisor to the company. The HSE Manager is responsible for developing compliance procedures, maintaining statutory logs, facilitating specialized environmental training, and conducting routine verification and compliance audits across all active projects.

4.2 The Environmental Aspects & Impacts Register

RAMA deploys a dynamic **Environmental Aspects and Impacts Register** tailored to each construction site. Project management teams must follow a strict sequential analysis when mobilizing to a new site to properly feed into this register.



The severity of any potential impact discovered during this evaluation is categorized into five distinct tiers:

1. **Negligible:** Localized incident with no environmental escalation or external footprint.
2. **Minor:** Contained, low-risk event (*e.g., a localized, minor site spillage on hardstanding*).
3. **Major:** Severe environmental breach (*e.g., chemical contamination of an adjacent watercourse*).
4. **Enforcement Notice:** Regulatory intervention carrying potential prosecution.
5. **Prohibition Notice:** Mandatory work stoppage resulting from a critical breach, with probable prosecution.

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT	Revision No. 02	Page 4 of 10

The live register actively tracks and mitigates these baseline environmental aspects across all projects:

- Energy consumption & carbon footprint
- Waste management, segregation, and legal disposal
- Water conservation and consumption limits
- Hazardous material spillages
- Fire risks and industrial emissions
- Dust generation and particulate tracking
- Acoustic pollution and vibration management

4.3 Employee Responsibilities

Environmental stewardship is an individual duty. Every RAMA employee is explicitly required to:

- Understand, respect, and comply with the specific site Environmental Aspects and Impacts Register.
- Handle, store, and transfer project construction materials strictly according to safety data sheets and training guidelines.
- Dispose of project waste only within designated streams; unmanaged waste exposes the company to severe regulatory penalties, remediation costs, and health hazards.
- Immediately report all environmental near-misses, anomalies, or active incidents (*such as chemical spills*) to the site HSE team.
- Proactively execute localized emergency containment procedures when an incident occurs.
- Eliminate wasteful practices to minimize the unnecessary consumption of water, electricity, and raw materials.

4.4 Key Operational Controls & Regulatory Alignment

All worksites must supplement these corporate policies with site-specific baseline conditions. Prior to breaking ground, teams should visually inspect all discharge routes, drainage infrastructure, collection ditches, and nearby natural water bodies.

Whenever accessible, a formal site drainage map must be filed to distinguish between surface water routing to containment interceptors, foul sewers, or direct discharges to natural rivers.

Statutory Mandate: In accordance with **Proclamation No. 295/2005 (Ethiopian Environmental Pollution Control)**, RAMA acknowledges that all enterprises and workplaces bear strict civil and legal liability for controlling, mitigating, and treating any form of pollutant generated during their operational activities.

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT	Revision No. 02	Page 5 of 10

4.5 Hazardous Liquid Storage & Fuel Management

- **Bulk Storage Secondary Containment:** All hazardous liquids, including fuel, lubricants, and chemical stores, must be housed within structurally sound bunded areas engineered to hold **110%** of the largest tank's maximum capacity. Storage must sit on reinforced, impermeable hard standing wherever operationally viable.
- **Authorized Refuelling Protocols:** Only trained, management-approved personnel are permitted to execute refuelling operations. Refuelling must take place exclusively at centrally bunded main tanks.
- **Remote Refuelling:** If field operations demand remote refuelling, it must occur at a designated distance away from surface drains and watercourses, remaining attended for the entirety of the transfer. Mobile bowzers must be securely locked post-use with all hoses reeled safely within the secondary skin. Drip trays must be strategically deployed to catch transient drips, backed by clear signage indicating the nearest spill response kit.

Static Diesel Tank Integrity Checklist

The following structural assessments must be verified and logged weekly:

- Is the bund wall entirely impervious, showing no structural cracks, shifting, or degradation?
- Does the containment volume successfully maintain its required 110% capacity margin?
- Are all ancillary fittings, valves, and distribution hoses housed completely within the safe boundary of the bund when idle?
- Is pipework arranged so that any pressurized pinhole leak cannot "overshoot" the bund walls?
- Are all fueling systems locked out securely at the conclusion of every shift to prevent unauthorized access?

Mobile Bowser & Drum Storage Standards

- **Bowzers:** All mobile units must be double-skinned (110% fluid capacity design), equipped with automatic shut-off refueling nozzles, and feature heavily secured sight glasses. Fluid levels between the double skins must be inspected frequently, with any captured fuel extracted immediately.
- **Drum Storage:** Drums or small containers of hazardous liquids (*e.g., hydraulic fluids*) must rest continuously inside dedicated drip trays capable of retaining a minimum of **25%** of the total volume of the largest container stored on it. individual drum storage under this provision is strictly capped at a maximum of **200 liters**.
- **Volatile Storage:** All paints, solvents, and highly volatile compounds must be locked within specialized, ventilated hazardous storage units.

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT	Revision No. 02	Page 6 of 10

4.6 Project Waste, Energy, & Resource Management

- **Site Waste Management Plans (SWMP):** Every major project must maintain a structured SWMP designed to reduce raw waste metrics. These plans must compile accurate volume records of ongoing waste streams while providing actionable forecasts for upcoming project phases.
- **Energy Conservation:** All machinery, plant, vehicles, and office equipment must be powered down completely when not actively in use. Security lighting should be calibrated to run on smart sensors or strictly monitored timers. Site offices must display prominent energy conservation signage, and water/power consumption data must be audited monthly against project baselines.
- **Material Lifecycles:** Materials suitable for repurposing or backfilling must be sorted and reused on-site wherever engineering specifications permit. Weather-sensitive supplies must be housed within covered, moisture-controlled storage structures to prevent material degradation.
- **Sustainable Infrastructure Evaluation:** Project planners must consider the long-term, whole-life environmental impacts of our works. Engineering reviews must actively plan for shifting environmental legislation, localized climate risk factors, raw resource depletion curves, material cost volatility, and the ultimate decommissioning impacts of the asset.

4.7 Emergency Control & Emergency Training

The Managing Director and the HSE team ensure that every RAMA project maintains a functional Emergency Response Plan specifically targeted at chemical, fuel, or sewage spills. In the event of a breach, fields teams must instantly follow the verified step-by-step containment protocols outlined below and file a formal incident report via the **RAMA Accident & Incident Reporting Procedure**.

To support this execution, RAMA delivers a multi-tiered environmental training matrix:

- **Environmental Inductions:** Mandatory for all newly onboarding staff, outlining core site sensitivities and individual duties.
- **Targeted Competency Briefings:** Dedicated technical sessions covering the live Environmental Aspects Register and its direct impact on specific engineering Method Statements.
- **Toolbox Talks:** Weekly, fast-paced field briefings focused on daily practical topics like safe waste segregation, spill kit deployment, and clean refueling.

4.8 Inspection, Audit, & Procurement

Active site controls must be verified through structured inspection schedules. The HSE Manager, often in collaboration with the client's environmental representatives, carries out independent compliance audits to verify adherence to legal frameworks, contractual obligations, and standard best practices.

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT	Revision No. 02	Page 7 of 10

If corrective interventions are flagged during an audit, the Managing Director reviews the structural requirements, authorizes the immediate execution of remedial works, and formally signs off on the compliance tracker to confirm closure.

Furthermore, RAMA actively expands its sphere of influence by requiring all supply chain partners, vendors, and subcontractors to pass basic environmental performance metrics during the procurement cycle.

5. Emergency Spill Response Protocols

Protocol A: Containment and Treatment of Spillages to Land

Steps	Controls	Absorbent Materials To be applied
1.	Health & Safety Controls Immediate Response Identify the spilled compound. Apply appropriate Control of Substances Hazardous to Health (COSHH) workflows. Don the required Personal Protective Equipment (PPE) and immediately eliminate all potential ignition sources in the immediate area.	
2.	Sensitive Area Isolation Containment Preparation Rapidly build a protective barrier using sand, inert materials, or dedicated absorbent socks around any vulnerable areas, surface channels, or drainage points down-gradient of the spill.	Sand or inert material
3.	Construct Flow Bunds Topographical Containment If the fluid release occurs on sloped terrain, construct an emergency containment bund across its path using earth-moving equipment (excavators/dozers) or manual tools. For minor flows, leverage localized absorbent booms. If time permits, line the containment basin with heavy plastic sheeting to stop soil infiltration.	Booms or sand Available material and plastic sheeting Available material and plastic sheeting

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT	Revision No. 02	Page 8 of 10

Steps	Controls	Absorbent Materials To be applied
4.	Source Remediation Mitigation Isolate and treat the primary source of the pollution by closing stuck valves, patching tank punctures, or uprighting compromised storage drums.	
5.	Apply Specialized Absorbents Cleanup Phase Formalize the boundaries of the impacted area. Apply appropriate media to bind the surface liquids. For Oils/Fuels , use white treated granules, sand, or specialized oil absorbent rolls. For Chemicals , apply dedicated chemical pads or sand. For Sewage , coat the zone with soil or sand. Gather used materials into heavy-duty hazardous waste bags and place them directly into designated hazardous waste skips.	<i>Oil Based</i> - Treated Granules - Oil absorbent rolls, Sand <i>Chemicals</i> - Chemical Pads - Sand <i>Sewage</i> - Soil, Sand
6.	Substrate Contamination Review Post-Incident Audit Excavate and inspect the underlying soil beneath the spill zone for signs of migration (noting changes in soil appearance, smell, or texturing). If residual contamination is suspected, execute formal soil core sampling. Remove all impacted substrate to a secure remediation area pending chemical treatment or authorized off-site disposal.	

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT	Revision No. 02	Page 9 of 10

Protocol B: Control and Treatment of Spillages to Water Resources

Steps	Controls	Absorbent Materials
1.	Health & Safety Controls Immediate Response Cross-reference the substance with COSHH data sheets, don appropriate marine/water-safe PPE, and secure the zone against potential ignition risks.	
2.	Waterway Inflow Prevention Critical Bank Isolation Construct robust earth bunds or position dense absorbent booms directly along the banks of nearby ditches and streams to prevent the land-based spill from introducing pollutants into the aquatic ecosystem.	Bunds Oil Absorbent Booms, Sand Soil
3.	Ditch Damming Dynamic Channel Isolation If a pollutant successfully enters an active drainage ditch, dam the channel downstream immediately using sandbags or local inert soils. If water flow is high, line the upstream face of the dam with plastic sheeting to establish a temporary impermeable water lock.	Sand Bags or local inert material.
4.	Floating Contaminant Collection Surface Remediation Deploy floating containment booms completely surrounding the surface slick to halt downstream migration. Use white oil-absorbent pads/rolls for petroleum products or black specialized pads for floating chemicals . Sweep and compress the booms across the surface to harvest the slick, continuing until zero surface sheen remains.	Booms Oils – white absorbent pads or rolls Chemical – black pads
5.	Sinking Chemicals Protocol Deepwater Escalation	

	Company Name RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-012	
Title	ENVIRONMENTAL GUIDANCE CRITERIA DOCUMENT		Revision No. 02 Page 10 of 10

Steps	Controls	Absorbent Materials
	If the substance sinks below the surface, execute all possible upstream source controls to limit further entry. Immediately escalate the incident to the HSE Manager and corporate officers to coordinate specialized underwater dredging, pumping, or chemical neutralization systems.	
6.	Reinstatement & Disposal Final Closure Once laboratory testing or visual audits confirm that all surface and structural contaminants are gone, carefully extract the deployed dams, booms, and pads. Package all saturated media into certified hazardous waste containers for controlled disposal. Remobilize the bank lines and stream bed back to their baseline conditions.	

5. Related Document

Document Title

Document No.

Environmental Management System Planning form	RAMA-OF-032
Environmental Daily Checklist	RAMA-OF-051
Environmental Management System Status Report	RAMA-OF-055
House Keeping Form	RAMA-OF-1002

6. DISTRIBUTION

- MD, GM, COO
- All Department
- All Projects

Approval Name:	Signature:	Date: 01/01/2020
--------------------------	------------	------------------